

Defense Brief — Representation Convergence Project

Companion to the 84-page report. One page of framing per topic; every number cited is measured and appears in the report with its generator notebook.

1 · Assumptions (what the results are conditional on)

Item-level correspondence exists and is known for supervised fits. Row *i* of every array is the same passage or image. Constructed (nested sorted-id prefixes, deterministic recipe) and verified numerically; C.12 deliberately removes it.

Linear maps, closed form. Every fitted map is ridge regression — no gradients, no iteration. The MLP contrast (F2) converged to the linear answer exactly.

Frozen encoders. No pretrained weight modified anywhere. F3 (unfreeze) built but gated shut to preserve this.

Held-out discipline includes transforms. Splits precede fitting; whitening covariances, hub bases, and alpha from training rows only. Gallery 1,000 everywhere, chance 0.001.

Geometry diagnostics trusted as measurements. Effective rank and mean pair-cosine treated as valid; C.14's perfect rank ordering is itself evidence.

2 · Restrictions (scope the claims do not exceed)

claim	measured scope	NOT claimed
convergence is linear	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 ×3; 4 text encoders	all architectures, all scales
alignment scales with capacity	DINOv2 22M–304M; E.16: giant + Qwen FALSIFIED	monotonic beyond 304M (both ladders fall)
isotropy rescue	GPT-2, two image scales (102.6% / 101.4%)	sufficiency — C.14 bounds it
unsupervised recovery	39.7% exact, 595× chance	a deployable pairing-free system
shared hub	7 encoders, 4 objectives, COCO; transfer 92.9–96.7%, cross-lineage and cross-architecture	a universal space for all models, domains, or languages
natural-world data matters	fractal R ≈ 0.09–0.21 (G9); transfer at floor (G10)	that ANY structured training fails (only FractalDB tested)

One language, one image domain, one caption corpus throughout. Every '% of ceiling' against a ceiling re-measured under the same protocol.

3 · Tuning (every knob, and how it was set)

knob	how set	guard
ridge alpha	val sweep 1e-3..10	chosen on train-internal, never eval
pooling	fixed cls+patch across sweep	one mixed run excluded
N (pairs)	scaled with width → 11.1 rows/dim	starvation biases DOWN
GW epsilon	swept per run; optimum moves	reported as sweep, not single value
HUB_DIM × alpha	2-D sweep, rows/dim ≥ 5	pre-registered; cliff reported as finding
gates	90% ceiling; 85% transitivity; ≥5 rows/dim	registered before results; 89.0% = NOT met

4 · Defense — the hard questions, with answers

"Isn't whitening cheating?" Label-free, fitted on training targets only, linear — an instance of the thesis, not an exception. The full map is whiten ■ ridge: still one matrix.

"Why is whitened R^2 0.014?" Whitening perfects angles, destroys magnitudes. R^2 asks about magnitudes; retrieval asks about angles. One system, two metrics, opposite verdicts.

"Isotropy helps — then why does it destroy C.12?" Reading vs finding. Cosine reads a KNOWN correspondence: uneven variance is noise. GW FINDS an unknown one: uneven variance is signal. Same property, opposite prescriptions — measured twice each.

"Why did the hub break at 768?" Least squares divided by a near-null singular value. The head's coordinates have algebraic rank 768, so at hub width 768 the last direction is σ 0.0054 vs 10.79 one step earlier. Seven accounts falsified first; the eighth predicted the location on two out-of-sample encoders. Truncation or $\alpha=1e3$ repairs it. A numerical artifact of the fit, not a property of shared coordinates.

"Is the transfer result circular?" The random-map control replaces only the hub map and collapses to chance (0.001). Absolute R@1 RISES on unseen encoders, which a task-memorising head cannot produce.

"Have the artifacts been checked?" 47 checks, NO hard failures. Every fit 5.9–9.8 rows/dim; shuffle gaps 0.545–0.863 vs 0.2 threshold. Two flags are the documented GPT-2 collapse. (Appendix D.5.)

"How do I know the numbers are right?" Five documented self-corrections via measurement, width-tagged checkpoints, transfer checks against known pairs, AST-validated notebooks. The error CLASS is real; the defenses are structural.

5 · Achievements — the nine findings

#	finding	headline evidence	section
1	convergence is linear	MLP = ridge exactly (0.9183 = 0.9183)	C.9/F
2	related, not rigid	Procrustes 0.038 vs ridge 0.592	\$4, B
3	deployable	95% of ceiling @ 0.79 MB; AUC 0.999	\$4, D
4	cross-modal, no joint training	40.9% of ceiling, 358× chance	C.8
5	alignment scales (bounded)	+15.3 pts/decade; E.16 FALSIFIED beyond	C.10, E.16
6	isotropy, not similarity training	whitening → 102.6% / 101.4% at two scales	C.11
7	correspondence is self-identifying	only 39.7% = 595× chance, 0 pairs	C.12
8	one shared space, used	hub 92.9–96.7% of native; ConvNeXt ABOVE the band	C.13
9	the mechanism predicts	rank-corr +1.00; isotropy necessary-not-sufficient	C.14

Plus the meta-finding, measured TWICE: isotropy helps reading a correspondence and hurts finding one (C.11 vs C.12). Eleven falsified explanations, two the project's own.

6 · The scale caveat

"Your text encoders top out at bge/BERT/GPT-2 scale (768–1024-d). Doesn't that put you in a different regime?" Yes, and worth conceding before it's raised. The effect is clearly measurable at this scale — the local-structure and objective-over-lineage findings hold on encoders 2–3 orders of magnitude smaller than PRH's LLMs. The closed-form, CPU-only design is the point.

"Bigger models converge more?" Tested it, pre-registered. DINOv2 ladder: ρ 0.381 → 0.381 → 0.418 → 0.339 at giant. Text ladder: 0.320 → 0.352 → 0.307 at Qwen-4B. Both rise then fall. Both pre-registered thresholds FALSIFIED. (E.16.)

"Percentages fall as encoders get bigger?" Absolute R@1 rises monotonically: 0.398, 0.442, 0.468. The percentage falls because the denominator (native head) rises faster. DINOv2-large's entry map at 4.2 rows/dim is below the floor of 5, so 92.9% is conservative.

7 · The target-choice objection

"Why bge-m3 when SBERT scores better?" Comparability, not merit. SBERT wins on both axes: writability 101.0% vs 98.1%, raw R@1 0.510 vs 0.466. bge retained because every ceiling is calibrated against it; switching would make figures incomparable for ~3 points. SBERT is the alternative in the demo notebook.

"bge-m3's dual role — hub member and target?" bge's text embeddings are the evaluation criterion, not a free parameter. The head maps hub → bge text and R@1 is scored against bge captions. This is a design choice disclosed throughout, not a confound — the random-map control proves the head alone carries nothing.

"SBERT is cleaner geometrically too?" It is — bge pair-cosine +0.57 vs SBERT +0.135. But that anisotropy is the CAUSE of the two gaps already on the table. Counting it separately double-counts one fact. And bge being the harder target makes every result MORE conservative: a cleaner SBERT ceiling would pull every reported percentage DOWN.

8 · Conclusion

The thesis was tested in every posture available: pairwise and supervised (holds, MLP adds nothing); across modalities without joint training (41–58% of ceiling, rising with capacity); with the metric geometry interrogated (apparent failures were coordinate-frame artifacts); with the pairing withheld (correspondence partially identifies itself from shape); and constructively (one 512-d hub serves seven encoders, a component trained on one transfers to encoders it never saw at 93–97% of native). Two things distinguish the project: its negatives are load-bearing and mechanistic, and its errors are part of the record.

9 · Post-defense: the fractal probe (G8–G10, Appendix F)

Added after the defense as ongoing research. Nothing in the main report depends on this section; it extends the 'well-trained' clause with a new control.

The question: does convergence need natural-world data? Two FractalDB-1k models (ResNet-50, DeiT-tiny — zero natural images) measured against all eight roster encoders.

G9 — visible geometry

prediction	CNN	ViT
P1 fractal < ImageNet twin	CONFIRMED 8/8	CONFIRMED 8/8
P2 fractal > random twin	CONFIRMED 4/4	CONFIRMED 4/4
P3 image-vs-text R gap	INCONCLUSIVE	INCONCLUSIVE
P5 fractal > pixels (raw)	FALSIFIED 0/4	CONFIRMED 4/4

Reality fraction $R \approx 0.09$ (CNN) / 0.21 (ViT). Whitened P5w CONFIRMED 4/4 ($R_{\text{white}} \approx 11\%$). Preprocessing confound controlled: norm accounts for $\sim 10\%$, training $\sim 90\%$.

G10 — hub recoverability

encoder	transfer	native	random twin	interpretation
Fractal CNN	0.014	0.014	0.012	at floor; = random
Fractal ViT	0.020	0.022	0.017	at floor; = random
ImageNet CNN	0.258	0.274	—	clears floor
ImageNet ViT	0.204	0.208	—	clears floor

H1 FALSIFIED both architectures. **H3 NOT MEANINGFUL** — both transfer and native at the 0.050 floor; %-of-native ratios blocked by the floor guard. **H2 CONFIRMED** — ImageNet twins transfer at 0.20–0.26.

What the fractal probe establishes

Fractal training creates real shared geometry above random (P2 confirmed), so convergence is learned rather than purely architectural. But that geometry does NOT transfer into semantic caption retrieval through the frozen hub — fractal encoders remain at the random floor while ImageNet twins transfer strongly. Natural-world experience is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.

Four implications for how alignment is measured

1. Check isotropy before believing a failure. 2. Visible $\neq$ recoverable — raw agreement does not bound what a fitted map can extract. 3. No single 'good geometry' — whitening's prescription inverts with the task. 4. Convergence is local, so the hypothesis holds in its narrow form, not its universal one.

Falsification ledger update: 11 (was 10). Entry #11: 'fractal geometry is recoverable through the hub' — FALSIFIED (G10/F.4). Two of the eleven are the project's own.